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Gemini 3 Deep Think: Advancing science, research and engineering

Feb 12, 2026

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Our most specialized reasoning mode is now updated to solve modern science, research and engineering challenges.

The Deep Think team

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Today, we're releasing a major upgrade to Gemini 3 Deep Think, our specialized reasoning mode, built to push the frontier of intelligence and solve modern challenges across science, research, and engineering.

We updated Gemini 3 Deep Think in close partnership with scientists and researchers to tackle tough research challenges — where problems often lack clear guardrails or a single correct solution and data is often messy or incomplete. By blending deep scientific knowledge with everyday engineering utility, Deep Think moves beyond abstract theory to drive practical applications.

The new Deep Think is now available in the Gemini app for Google AI Ultra subscribers and, for the first time, we're also making Deep Think available via the Gemini API to select researchers, engineers and enterprises. Express interest in early access here.

Here is how our early testers are already using the latest Deep Think:

Lisa Carbone, a mathematician at Rutgers University, works on the mathematical structures required by the high-energy physics community to bridge the gap between Einstein's theory of gravity and quantum mechanics. In a

field with very little existing training data, she used Deep Think to review a highly technical mathematics paper. Deep Think successfully identified a subtle logical flaw that had previously passed through human peer review unnoticed.

At Duke University, the Wang Lab utilized Deep Think to optimize fabrication methods for complex crystal growth for the potential discovery of semiconductor materials. Deep Think successfully designed a recipe for growing thin films larger than 100 µm, meeting a precise target that previous methods had challenges to hit.

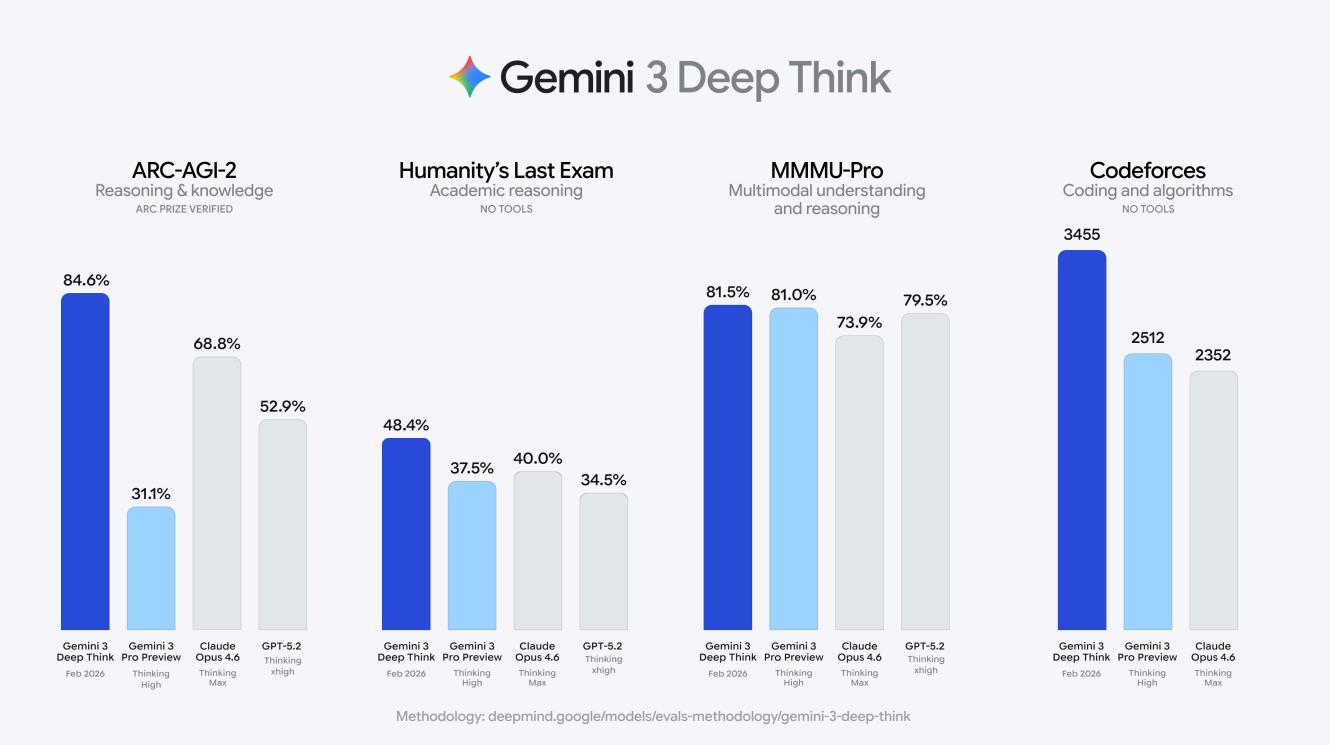
Anupam Pathak, an R&D lead in Google's Platforms and Devices division and former CEO of Liftware, tested the new Deep Think to accelerate the design of physical components.

Elevating reasoning with mathematical and algorithmic rigor

Last year, we showed that specialized versions of Deep Think could successfully navigate some of the toughest challenges in reasoning, achieving gold-medal standards at math and programming world championships. More recently, Deep Think has enabled specialized agents to conduct research-level mathematics exploration.

The updated Deep Think mode continues to push the frontiers of intelligence, reaching new heights across the most rigorous academic benchmarks, including:

- Setting a new standard (48.4%, without tools) on Humanity's Last Exam, a benchmark designed to test the limits of modern frontier models
- Achieving an unprecedented 84.6% on ARC-AGI-2, verified by the ARC Prize Foundation
- Attaining a staggering Elo of 3455 on Codeforces, a benchmark consisting of competitive programming challenges
- Reaching gold-medal level performance on the International Math Olympiad 2025



Navigating complex scientific domains

Beyond mathematics and competitive coding, Gemini 3 Deep Think now also excels across broad scientific domains such as chemistry and physics. Our updated Deep Think mode demonstrates gold medal-level results on the written sections of the 2025 International Physics Olympiad and Chemistry Olympiad. It also demonstrates proficiency in advanced theoretical physics, achieving a score of 50.5% on CMT-Benchmark.

Benchmark		Gemini 3 Deep Think Feb 2026	Gemini 3 Pro Preview Thinking High	Claude Opus 4.6 Thinking Max	GPT-5.2 Thinking xhigh
ARC-AGI-2 Abstract reasoning puzzles	ARC prize verified	**84.6%**	31.1%	68.8%	52.9%
Humanity's Last Exam Academic reasoning (full set, text + MM)	No tools Search + code execution	**48.4%** **53.4%**	37.5% 45.8%	40.0% 53.1%	34.5% 45.5%
MMMU-Pro Multimodal understanding and reasoning	No tools	**81.5%**	81.0%	73.9%	79.5%
International Math Olympiad 2025 Mathematics		**81.5%**	14.3%	-	71.4%
Codeforces Coding and algorithms	No tools, Elo	**3455**	2512	2352	-
International Physics Olympiad 2025 (theory) Physics		**87.7%**	76.3%	71.6%	70.5%
CMT-Benchmark Condensed matter theory		**50.5%**	39.5%	17.1%	41.0%
International Chemistry Olympiad 2025, (theory) Chemistry		**82.8%**	69.6%	-	72.0%

Methodology: deepmind.google/models/evals-methodology/gemini-3-deep-think

Accelerating real-world engineering

In addition to its state-of-the-art performance, Deep Think is built to drive practical applications, enabling researchers to interpret complex data, and engineers to model physical systems through code. Most importantly, we are working to bring Deep Think to researchers and practitioners where they need it most — beginning with surfaces such as the Gemini API.

With the updated Deep Think, you can turn a sketch into a 3D-printable reality. Deep Think analyzes the drawing, models the complex shape and generates a file to create the physical object with 3D printing.

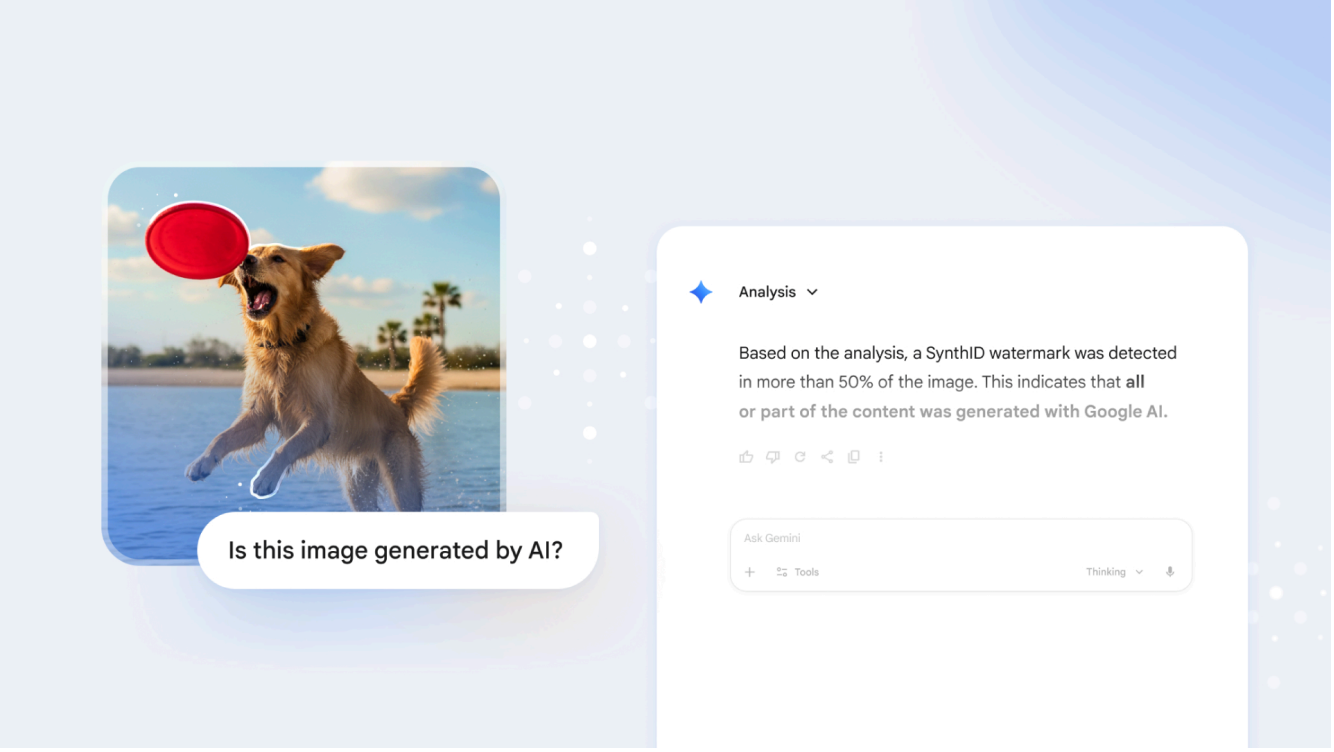
Available to Google AI Ultra Subscribers and the Gemini API via our Early Access Program

Google AI Ultra subscribers will be able to access the updated Deep Think mode starting today in the Gemini app. Scientists, engineers and enterprises can also now express interest in our early access program to test Deep Think via the Gemini API.

We can't wait to see what you discover.

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